Graphing exponential functions

We want to be able to graph **exponential functions**, which are functions in which the variable is in the exponent. These are all exponential functions:

$$f(x) = 3^x$$

$$f(x) = 2\left(\frac{1}{3}\right)^{-x+2} - 4$$

$$f(x) = -3^{x-1} - 2$$

$$f(x) = 6 \cdot 2^{-x} + 1$$

Characteristics of the graph

To learn how to graph exponential functions, let's start by graphing $f(x) = ab^x$, where a is a nonzero number and b is a positive real number not equal to 1.

The coefficient a is the **initial value**, because $f = a$ when $x = 0$, which means the y-intercept of an exponential function $f(x) = ab^x$ is the point $(x, y) = (0, a)$.

The base b is the **growth factor**.

- When $b > 1$, the function grows at rate proportional to its size.
- When $0 < b < 1$, the function decays at a rate proportional to its size.

The base has to be positive in order to ensure that the function will have a real-number output. For instance, if $b = -16$ and $x = 1/2$, then $f(x) = ab^x$ becomes

© Krista King Math



$$f\left(\frac{1}{2}\right) = (-16)^{\frac{1}{2}} = \sqrt{-16}$$

and we can't take the square root of a negative number. The base also can't be equal to 1 because $f(x) = a(1^x)$ gives $f(x) = a$ for all values of x, and then the function is no longer an exponential, it's a constant function.

The domain of the exponential function is $(-\infty, \infty)$, and the range is all positive real numbers $(0,\infty)$ if $a > 0$ and all negative real numbers $(-\infty,0)$ if $a < 0$.

Let's do an example so that we can see how to graph a simple exponential function.

Example

Graph the exponential function.

$$f(x) = 2^x$$

If we rewrite the function as $f(x) = 1(2^x)$, then we can identify the initial value $a = 1$ and the growth factor $b = 2$.

The initial value tells us that the function passes through $(0,1)$. And since $b > 1$, we know the function is increasing above the horizontal asymptote at $y = 0$.

We'll plug in a couple more values of x for which the value of $f(x)$ will be easy to calculate.

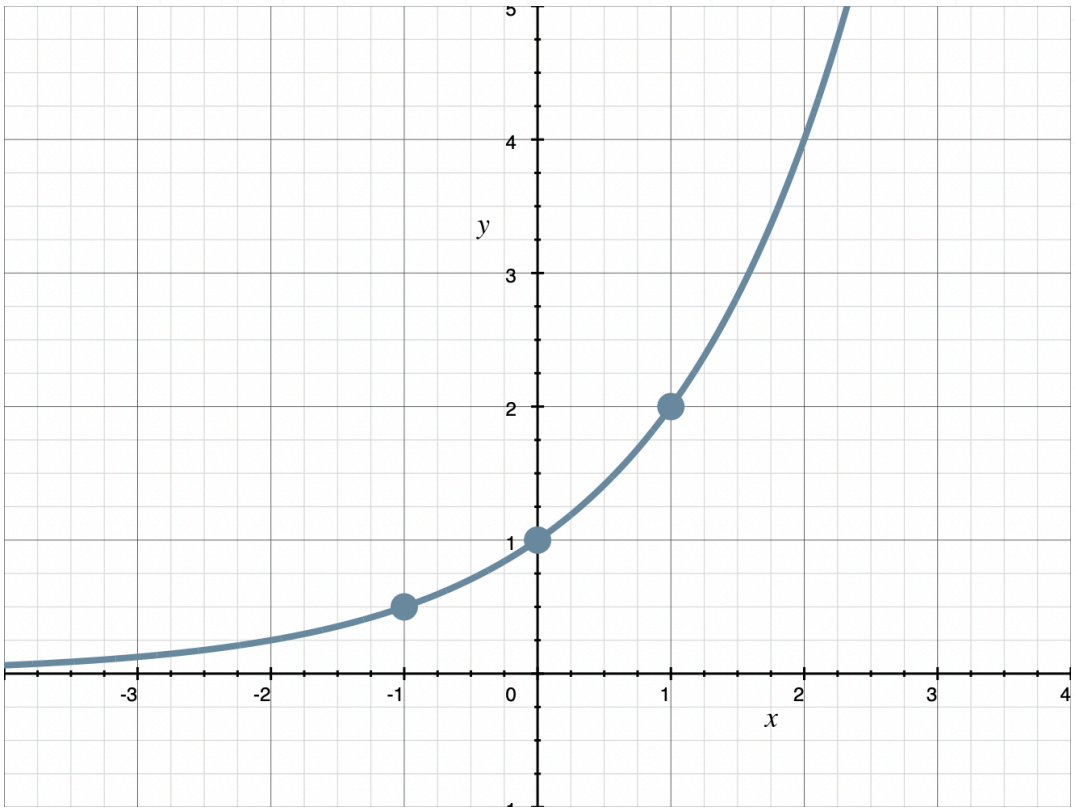
© Krista King Math



For $x = -1, f(-1) = 2^{-1} = 1/2$

For $x = 1, f(1) = 2^1 = 2$

Now we have three points on the graph of f: the y-intercept $(0,1)$, and $(-1,1/2)$ and $(1,2)$. If we plot these points and connect them with a smooth curve, we get



Asymptotes

Notice in the previous example that as we move to the left along the curve, the curve approaches the $x-$axis without passing through it. This is because the $y-$values of the curve get closer and closer to 0 ($y \to 0$), as x becomes more negative ($x \to -\infty$). So, we say that the line $y = 0$ is a horizontal asymptote of the function.

© Krista King Math

Generally speaking if k is a real number, and if $y \to k$ as $x \to \pm\infty$, then we have a **horizontal asymptote** at $y = k$. Conversely, if h is a real number, and if $x \to h$ as $y \to \pm\infty$, then we have a **vertical asymptote** at $x = h$.

When the exponential function is written as a function of x solved for y, the graph will have a horizontal asymptote, but the exponential function is written as a function of y solved for x, the graph will have a vertical asymptote.

Let's look at an example where we identify the asymptote from the equation.

Example

Will the graph of the function have a vertical asymptote or a horizontal asymptote? Find its equation.

$$x = -4^{y+3} - 2$$

Because the function is an exponential function that's solved for x in terms of y, the graph will have a vertical asymptote. To find it, we need to see what happens to x as $y \to \pm\infty$. To do this, we can plug in both $y = 100$ and $y = -100$ into the equation.

For $y = 100$,

$$x = -4^{100+3} - 2$$

$$x = -4^{103} - 2$$

© Krista King Math



$$x = -\text{(a very large number)} - 2$$

$$x \to -\infty - 2$$

$$x \to -\infty$$

Since x doesn't approach a real number, we don't have an asymptote here.

For $y = -100$,

$$x = -4^{-100+3} - 2$$

$$x = -4^{-97} - 2$$

$$x = -\frac{1}{4^{97}} - 2$$

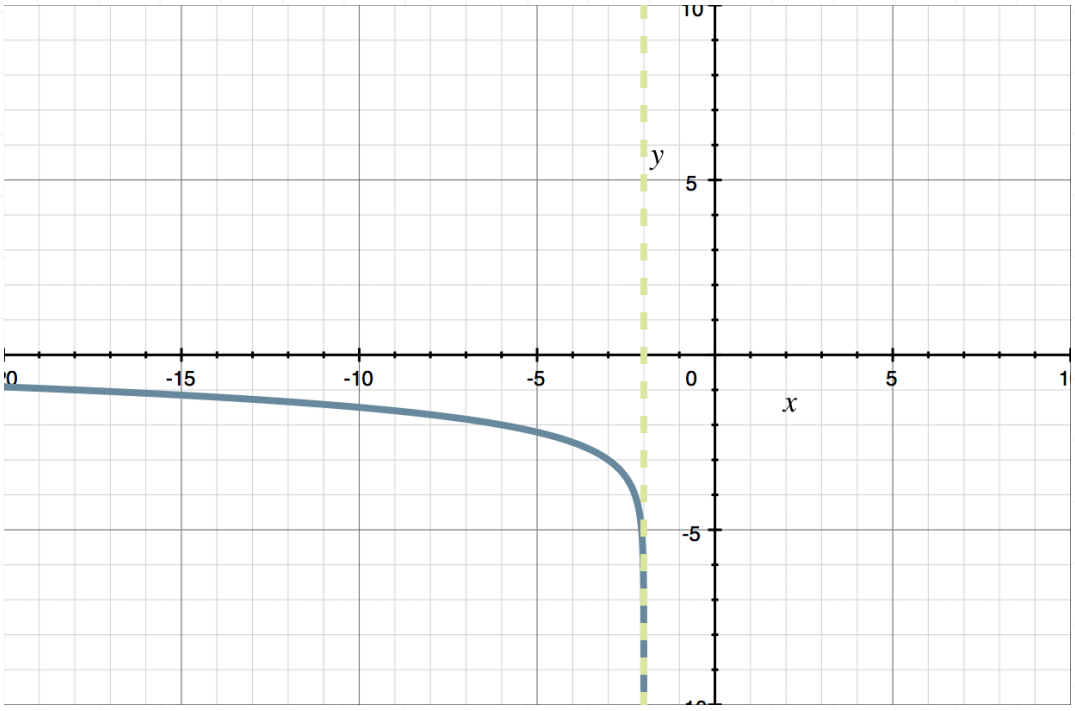
$$x = -\frac{1}{\text{a very large number}} - 2$$

$$x \to 0 - 2$$

$$x \to -2$$

So the vertical asymptote of the function is the line $x = -2$. A sketch of the function shows this result.

© Krista King Math



© Krista King Math